MANIPAL HACKATHON 2025

CODE FOR COEXISTENCE

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NIPPON KOEI INDIA PVT. LTD.
Consulting Engineers

ROUND 2

PROBLEM STATEMENT

GENERATIVE AI

AI-Powered Operations Co-Pilot for Mining and Infrastructure

Background

Mining and infrastructure projects across regions like India, Australia, UAE, Africa, and Europe generate vast amounts of operational data — covering equipment deployment, production, quality, fuel consumption, and maintenance. Traditional dashboards often make it slow and difficult for managers to extract meaningful insights from this data.

Challenge

The challenge is to build an AI-powered Operations Co-Pilot — a conversational assistant that understands natural language queries, retrieves relevant information from multiple data sources, and presents it in a clear and actionable manner. This will simplify access to operational insights and empower managers to make faster, better-informed decisions.

Features and Tools expected:

- Integration with LLMs for query understanding
- Use of Vector Databases for semantic search
- Data handling with MySQL / AWS Timestream
- Deployment on Web (Angular) and Mobile (Flutter)
- Clean, intuitive, and visually engaging UI/UX
- Ability to handle large-scale datasets efficiently

Relevant SDGs

- SDG 9: Industry, Innovation, and Infrastructure
- SDG 12: Responsible Consumption and Production
- SDG 13: Climate Action

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